

MITANSHU GOEL

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Education

Maharaja Agrasen Institute of Technology (MAIT), Delhi

B.Tech, Electronics & Communication Engineering; Minor in AI & Machine Learning

2022 – 2026

CGPA: 8.01 / 10

Experience

Nferent AI

Physical AI Intern

Mar 2026 – Jun 2026

Gurugram, India

- Wrote a 125 Hz C++ loop to steer two Elite CS66 arms from Quest 3 hand controllers. [Demo](#).
- Extended the loop to a Franka FR3 and recorded a 10-task LeRobot dataset across both platforms, with limits on workspace and hand speed. [Demo](#).
- Fine-tuned a $\pi_{0.5}$ vision-language-action policy on those 10 tasks and worked on its inference path.
- Built a MediaPipe vision classifier that lets a 20-motor Tesollo hand play rock-paper-scissors from RealSense video.

SarthakAI

AI Intern

Jun 2025 – Aug 2025

Delhi, India

- Built object detection for a UBTech Yanshee humanoid: a custom-trained YOLOv8 model on its camera stream.
- Wrote an OpenCV pipeline for gesture control, so a person can direct the same humanoid with hand signs.
- Added NeMo voice control so the humanoid answers questions or scans QR codes after a wake word. [Demo](#).
- Built a pick-and-sort line where a camera guides a robot arm to sort objects on a conveyor by colour, shape and type.

NextUp Robotics

Robotics Intern

Jul 2024 – Sep 2024

Ghaziabad, India

- Prepared a supplied 6-DOF arm's CAD-exported URDF for ROS 2, then checked its joints in Gazebo and RViz.
- Fixed URDF mismatches that broke arm trajectories, then ran MoveIt paths on real hardware. [Demo](#).

Projects

Quadruped, does warning a walking robot before a shove help? | *MuJoCo, PPO, Unitree Go1*

[Demo](#) | [Code](#)

- On mjlabs' Unitree Go1 walking task, added a shove every few seconds and a warning of its force and timing.
- Trained two PPO policies that differ only in whether they see the warning, 393M steps each, 4096 simulated robots.
- Without its warning at test time, the same policy fell on 52.3% of 250 N shoves, versus 1.2% with warning.

Hexapod, 18-DOF walking robot (A.T.O.M. team project) | *ROS 2, ros2_control, Gazebo, Docker*

[Demo](#) | [Code](#)

- Wrote the ROS 2 node that turns a velocity command into a tripod walking gait, in Gazebo and on the real robot.
- Wrote closed-form inverse kinematics that turns each foot target into the coxa, femur and tibia angles of a leg.
- Put the ROS 2 and Gazebo setup in Docker so the whole team worked from one environment.

Technical Skills

Robot Learning & Simulation: PyTorch, LeRobot, PPO, MuJoCo, mjlabs

Robotics: ROS 2, ros2_control, MoveIt, URDF, Gazebo, RViz, rosbridge

Perception: YOLOv8, OpenCV, MediaPipe, NVIDIA NeMo

Languages & Tools: C++, Python, Docker

Positions of Responsibility

Core Member, A.T.O.M. Robotics Lab, MAIT | *student robotics society*

Oct 2023 – Present

- Build the team's robots and represent MAIT at competitions, including Robocon, Flipkart GRID and IRDC.
- Built a browser page that drives a robot arm over a rosbridge WebSocket, with its camera feed live.